

Thesis Talk

Salome Bachmann

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Overview of Topics

- Context
- Terminology
- Overview of Models
- Results of Cohesive Zone Model
- Discussion
- Outlook

What am I doing and why is this relevant?

- Hang out in my office and have coffee (please come by)
- At the beginning, oversample and knit an entire top because the run time was too long
- Trying to make an accurate model for crack propagation in snow using SALVUS (FE)
- Cracks in snow on large scales can lead to avalanche release
- Average of 24 deaths by avalanche in Switzerland per year, multiple hundred non-fatal accidents (SLF, 2025)
- Understanding relevant because avalanches endanger human life and infrastructure (Schweizer et al., 2003)
- **Ultimate Goal:** determining whether slab breakoff can be seen in DAS data and how this would look

Words I will probably say a lot explained I

- Slab: Part of the snowpack which breaks off and slides downhill as a cohesive unit (the avalanche itself, I'm looking at dry-snow slab avalanches) (Schweizer et al., 2003)
- Weak layer: Layer with different mechanical properties than the rest of the snow, cracks usually propagate in this layer because it's weaker than the rest of the snowpack (Schweizer et al., 2003)
- Mode I, Mode II and Mode III cracks: cracks opening as a response to stress in different directions:

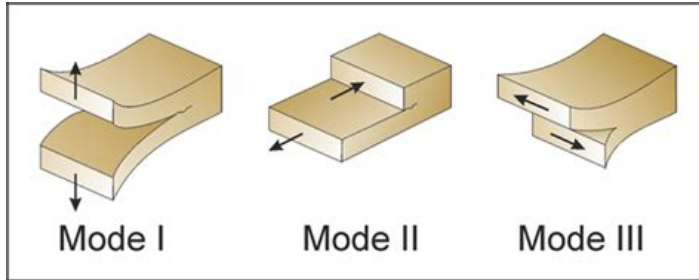


Figure: Fracture modes according to Philipp et al., 2013

Words I will probably say a lot explained II

- Anticrack: weak layer collapses under loading, slab starts bending, collapse and cracks caused by bending propagate (Gaume et al., 2018)
- Sub-Rayleigh: Cracks that propagate at speeds lower than the shear wave speed in snow (Trottet et al., 2022)
- Supershear: Cracks propagate faster than shear wave speed in snow (Trottet et al., 2022)

Overview of Models

- Model domain 400m wide, 3m high
- Absorbing boundary conditions 30m thick along top and both sides of domain
- No ABC at bottom because of theoretical snow interface
- 1.5m snow layer, 1.5m air layer
- Multiple source versions: single source, source line with sources firing with a time delay (moving source) and rupture front
- Sources buried in snow (like crack)
- For moving source: sub-rayleigh and supershear speeds
- 10Hz central frequency ricker source wavelet

My latest creation: the cohesive zone model (CZM)

- Model based on the cohesive zone model by Mahajan and Joshi, 2008
- In CZM, fracturing is a gradual process, separation of the crack faces takes place over an extended crack tip, the cohesive zone (Mahajan & Joshi, 2008)
- Resistance to opening by cohesive traction (Mahajan & Joshi, 2008)
- In other words: there is a zone ahead of the crack tip, which already has some damage of the bonds
- Cracks represented as a fraction between different crack modes (Mode I-III)
- Caused by a fault zone: starts at $x=30\text{m}$ and ends at $x=270\text{m}$, 0.5m depth (middle of snow layer)
- Sources every 10cm firing with a time delay (dependend on supershear vs sub-rayleigh)

Supershear vs Sub-Rayleigh Space-Time Plots - Velocity

Seismic waterfall: CZM sub-Rayleigh vs CZM supershear Horizontal particle velocity halfway in snow

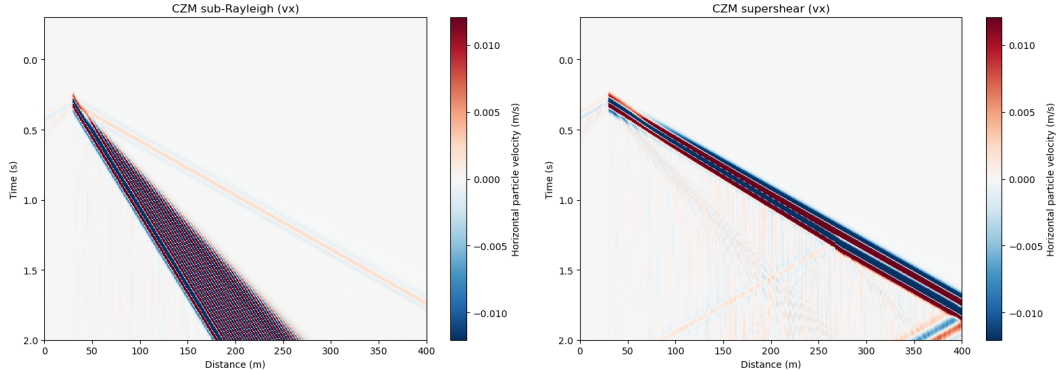


Figure: Horizontal Particle Velocity (v_x) in sub-rayleigh (left) and supershear case (right)

Supershear vs Sub-Rayleigh Space-Time Plots - Strain

Seismic waterfall comparison (strain (vertical component)): CZM sub-Rayleigh vs CZM supershear at mid-snow depth

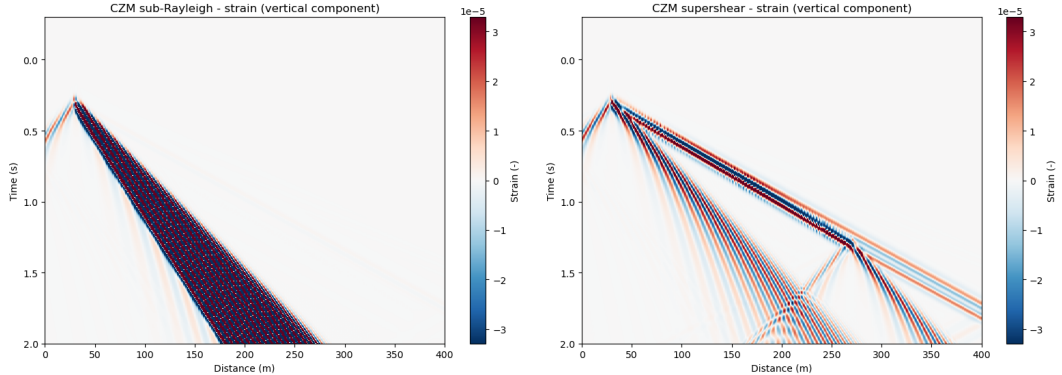


Figure: Strain in sub-rayleigh (left) and supershear case (right)

Possible interpretations of space-time plots I

- Sub-Rayleigh and Supershear case look very different (yay!)
- Blue-red-blue arrival before the main wavefront could be flexural wave (bending of the slab near the source)
- Blue first: negative velocity $\rightarrow$ source in negative y (slab is pushed down by the source)
- Red next: because of elastic rebound of slab
- Blue again: oscillation in source direction after rebound
- This can be seen in v_x for sub-rayleigh and strain for supershear
- In sub-rayleigh, waves travel slower than the flexural waves, which is very noticeable when looking at v_x
- In supershear v_x , rupture arrives with flexural waves
- Flexural waves can be seen in strain supershear plot: because strain is spatial derivative, which acts as a high-pass and amplifies flexural waves

Possible interpretations of space-time plots II

- This is because the derivative acts as a multiplication with $j\omega$, which is dependent on frequency: the higher the frequency, the greater $|j\omega|$, which means there is amplification of high frequencies (Oppenheim et al., 1997)
- Supershear case: Strong first arrivals, weak arrivals at later times
- "Reflection" seen in bottom right corner is rupture arrest front (energy could increase because rupture stops at a point)
- This is only visible in supershear case because there is constructive interference between the rupture front and the wavefronts because they move at similar speeds in supershear

Flexural wave -> early warning system?

- Because flexural wave arrives before main wavefront of cracks: could be used as early warning system
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Much needed sanity checks I

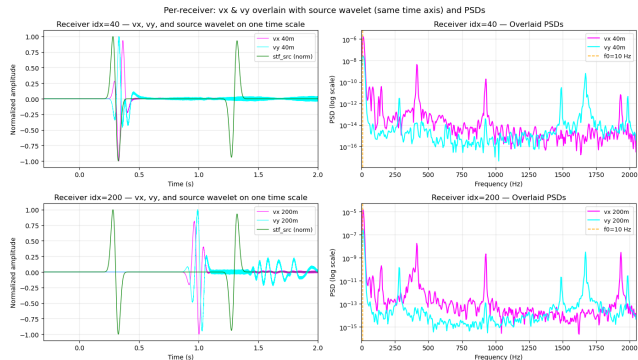


Figure: Traces and power spectral density (PSD) at receivers at 40m and 200m. Vertical velocity in blue, horizontal velocity in pink, source trace in green

Much needed sanity checks II

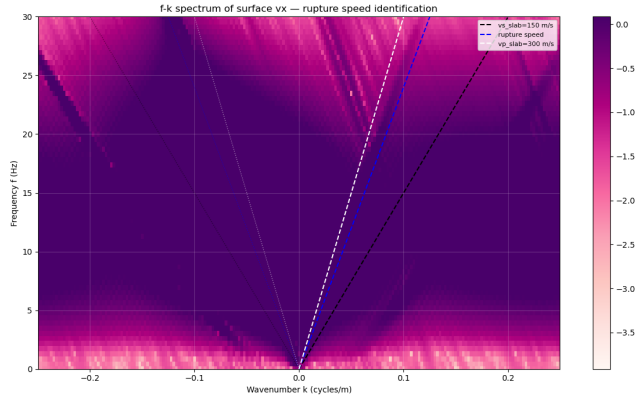


Figure: F-k plot for supershear case. Rupture speed shown by blue line, vp by white line and vs by black line.

Was it all just a dream?

- Source wavelet pulse (green) matches traces at both receiver locations
- Far trace (200m) has stronger coda noise in v_y : could be rupture arrest?
- All PSD have a clear peak at the source frequency (10Hz)
- F-k plot shows high energies at same slope of rupture speed: model takes the correct speed
- Trailing (s) wave can also be seen (line with similar slope to black line)
- Waves propagate in both directions (in rupture direction as well as against it)
- Probably not a dream, my model might be correct!

Next Steps

- Add weak layer to model
- Add topography to model
- Make an anticrack model using the Material Point Method (MPM)
- ???

Thank you for your attention! Any Questions or Suggestions?

Bibliography I



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Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS